



JC4002 Knowledge Representation

Day 4: Blocks World

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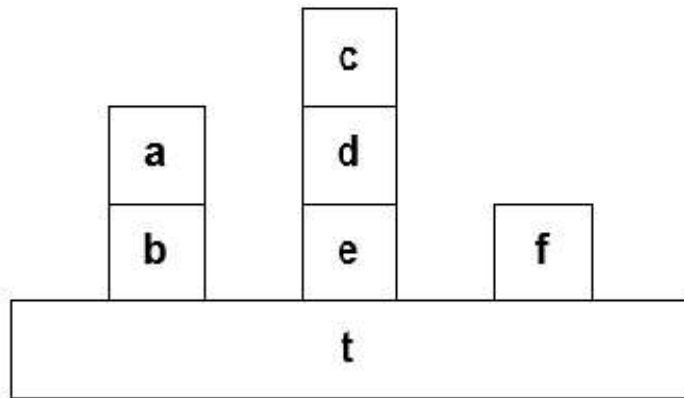
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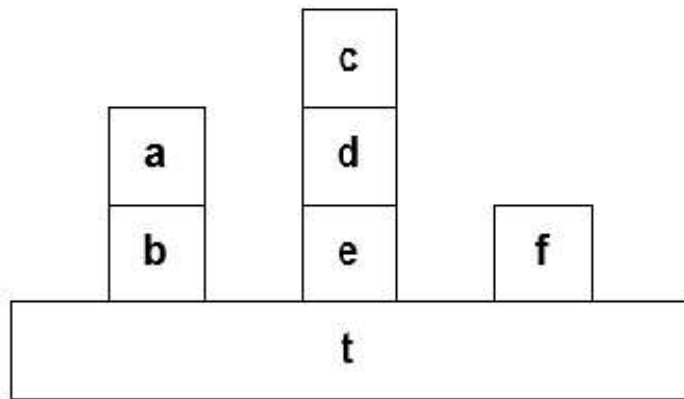
Blocks World

1. Problem Domain
2. Query Examples

Blocks World in Prolog



Blocks World in Prolog



```
1  block(a).
2  block(b).
3  block(c).
4  block(d).
5  block(e).
6  block(f).
7  ontop(a,b).
8  ontop(b,t).
9  ontop(c,d).
10 ontop(d,e).
11 ontop(e,t).
12 ontop(f,t).
13
14 free(t).
15 free(X) :- block(X), \+ ontop(_,X).
16
17 above(X,Y) :- ontop(X,Y).
18 above(X,Y) :- ontop(X,Z), above(Z,Y).
```

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Write the following queries and for each query describe (1) the query executed, and (2) the results.

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 1. | ?- block(X), above(X, t).
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Blocks World Queries II

- Are there blocks that are not on top of the table?
 1. | ?- block(X), \+ ontop(X, t).
 2. | ?- X = a; X = c; X = d; false.
- Which blocks are not free?
 1. | ?- block(X), \+ free(X).
 2. | ?- X = b; X = d; X = e; false.
- Is block a above the table?

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- Are there free blocks above a?
 1. | ?- block(X), above(X, a), free (X).
 2. | ?- false.
- Are there free blocks on top of the table?

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 2. | ?- true; false.
- Are there free blocks above a?
 1. | ?- block(X), above(X, a), free (X).
 2. | ?- false.
- Are there free blocks on top of the table?
 1. | ?- block(X), ontop(X, t), free(X).
 2. | ?- X = f.

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- Is the table free?
 1. | ?- free(t).
 2. | ?- true; false.
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- Is the table free?
 1. | ?- free(t).
 2. | ?- true; false.
- Is block a on top of itself?
 1. | ?- block(a), ontop(a, a).
 2. | ?- false.
- Which blocks are below block c?

Blocks World Queries III

- Is the table free?
 1. | ?- free(t).
 2. | ?- true; false.
- Is block a on top of itself?
 1. | ?- block(a), ontop(a, a).
 2. | ?- false.
- Which blocks are below block c?
 1. | ?- block(c), block(X), above(c, X).
 2. | ?- X = d; X = e; false.
- Which blocks are free and above block c?

Blocks World Queries III

- Is the table free?
 1. | ?- free(t).
 2. | ?- true; false.
- Is block a on top of itself?
 1. | ?- block(a), ontop(a, a).
 2. | ?- false.
- Which blocks are below block c?
 1. | ?- block(c), block(X), above(c, X).
 2. | ?- X = d; X = e; false.
- Which blocks are free and above block c?
 1. | ?- block(X), free(X), above(X, c).
 2. | ?- false.
- Is block d free?

Blocks World Queries III

- Is the table free?
 1. | ?- free(t).
 2. | ?- true; false.
- Is block a on top of itself?
 1. | ?- block(a), ontop(a, a).
 2. | ?- false.
- Which blocks are below block c?
 1. | ?- block(c), block(X), above(c, X).
 2. | ?- X = d; X = e; false.
- Which blocks are free and above block c?
 1. | ?- block(X), free(X), above(X, c).
 2. | ?- false.
- Is block d free?
 1. | ?- free(d).
 2. | ?- false.

Summary

- Example of blocks world in Prolog
- Asking queries and getting queries results

For today practical:

- You should have installed SWI-Prolog to your computer.